

Competitive Intelligence Valuation & Market Signal Prioritization

Competitive Intelligence Valuation & ROI/Cost Analysis · Client: Fortune 500 Medical Device Manufacturer

*The Problem: Reviewers had to manually screen thousands of new clinical trial filings a month to catch **competitor moves and emerging market signals**, a process too slow, inconsistent, and costly to keep up with a market that changes every week.*

Project Intro

- 1

Data Sourcing & Integration

Engineered a Python ingestion pipeline pulling 20,000+ public trial records from a national registry API on a weekly refresh, with automated sponsor deduplication and change tracking across 10+ relational tables.
- 2

Risk & Relevance Scoring Model

Designed a rule-based relevance score from sponsor, intervention, and condition signals, paired with an LLM-based importance score assessing trial novelty and market opportunity.
- 3

Model Validation & Risk Controls

Stress-tested model outputs with sensitivity and robustness checks, and built a guardrail layer that catches and blocks unsupported scores before they reach stakeholders.
- 4

Executive Valuation Reporting

Packaged results into a dashboard, quantifying competitive intelligence value for sponsor decision-making.

Impact

\$2M+ Sponsor-estimated intelligence value	6 Key competitors tracked
20K+ Trial records processed	3+ mo Manual review replaced
10+ Relational tables integrated	0 Unsupported scores shipped